Logos link to different BioCompute resources

BioCompute Objects | Galaxy PROJECT | HIVE | IEEE Advancing Technology for Humanity | CANCER GENOMICS CLOUD SEVEN BRIDGES

Publication of BioCompute Objects (IEEE-2791-2020) created from Galaxy workflow invocations

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Galaxy PROJECT | THE GEORGE WASHINGTON UNIVERSITY WASHINGTON, DC | BioCompute Objects

Solution: BioCompute

IEEE approved standard for communicating bioinformatic analysis workflows

- Acts like an envelope for entire pipeline
  - Can incorporate other standards
- Human and machine readable
  - Written in JSON
- Categorized by domains
- Adheres to and encourages F.A.I.R. principles
  - Fully open source
- Adaptable
  - e.g. to other schemas
- Preserves data provenance
- Unique IDs for versioning

Introduction to BioCompute

Key Features of a BCO

- Abstract away workflow based on commonalities
  - Platform/tool/protocol independent
- Usability Domain
  - Free text description of scientific purpose
- Data provenance
  - Data manifest, track process from beginning to end
  - Track user attribution ("authoredBy," "contributedBy," "reviewedBy," etc.)
- Verification Kit
  - Error Domain + IO Domain
  - Sanity check: given the input files and the inherent (or known) error, is the output this analysis claims to have gotten valid?
- Extensible
  - Extension Domain
  - Open source repository
- Embargo Field
  - Prevent others from viewing a BCO for any amount of time

BioCompute participants

Introduction to BioCompute

Metadata

Provenance Domain | Execution Domain | Description Domain | IO Domain | Error Domain | Extension Domain | Parametric Domain | Usability Domain

Introduction to BioCompute

Solution: BioCompute

Experimental Design | Analysis Steps | Parameters | Error | Data Source | Results | Attribution | Standardizes report content

BioCompute streamlines reporting without enforcing any tool, platform, or workflow strategy.

Machine readability enables customized views

1. User registers account on Portal
2. User gets API key from account
3. User creates a draft BCO from a Galaxy invocation
6. User adds information not automatically generated by Galaxy

Acknowledgements and Contact

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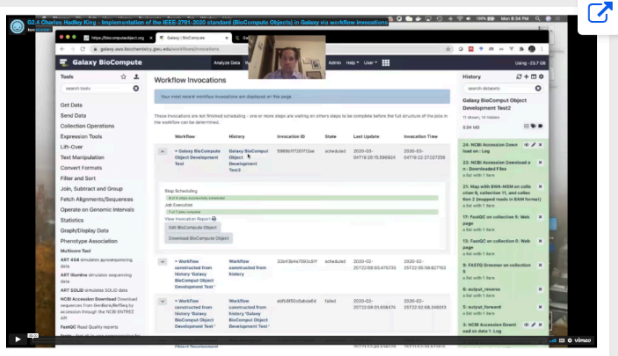
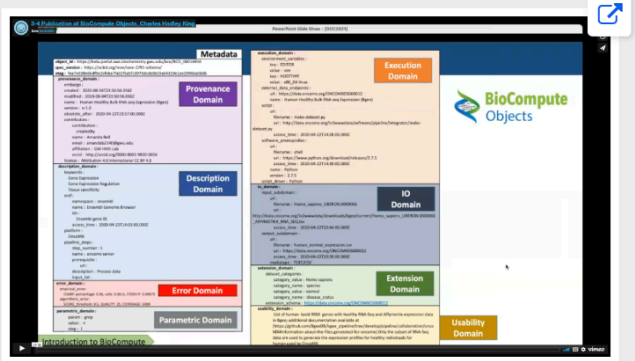
GCC 2020



BETA BCO Portal

BioCompute Objects



GCC 2021